

Supplementary Online Content

de Boer IH, Zelnick LR, Ruzinski J, et al. Effect of vitamin D and omega-3 fatty acid supplementation on kidney function in patients with type 2 diabetes: a randomized clinical trial [published November 8, 2019]. *JAMA*. doi:10.1001/jama.2019.17380

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This supplementary material has been provided by the authors to give readers additional information about their work.

eAppendix. Supplementary Methods

Laboratory drift and shift due to changes in reagents, calibrator lots, equipment, or equipment settings can introduce selective or nonselective bias into analyses.^{1,2} Therefore, we prospectively incorporated 5 levels of quality control (QC) into our measurements of creatinine, cystatin C, and urinary albumin in order to prevent lab error and, if present, to effectively detect and account for changes over time.³

First, assay controls provided by reagent manufacturers were run daily prior to clinical samples to verify proper analyzer function. Second, we generated aliquots of “high” and “low” serum and urine standards, prepared by the hundreds and frozen at -80°C, and measured them daily in parallel with clinical samples to monitor for drift.

Third, we generated a “QC set” of 20 serum samples and a parallel “QC set” of 20 urine samples in order to harmonize laboratory data across the study if shift or drift was detected. For the serum QC set, the 20 samples were selected to span large ranges of serum creatinine and cystatin C concentrations. The urine QC set of 20 samples was similarly designed to span large ranges of urine creatinine and albumin concentrations. The samples for each QC set were aliquoted and frozen at -80°C and measured throughout the study. Samples of the serum and urine QC sets were each measured a minimum of 8 times over the course of sample analysis.

Fourth, blind replicates (5-10%) were analyzed to ensure the integrity of the entire assay and data management process, including the identification and labeling of samples and the incorporation of results into master databases. Fifth, national and international standards were

measured to verify that serum measurements of creatinine and cystatin C were consistent with National Institute of Standards and Technology (NIST) and European Reference Materials (ERM-DA471/IFCC) standards, respectively.

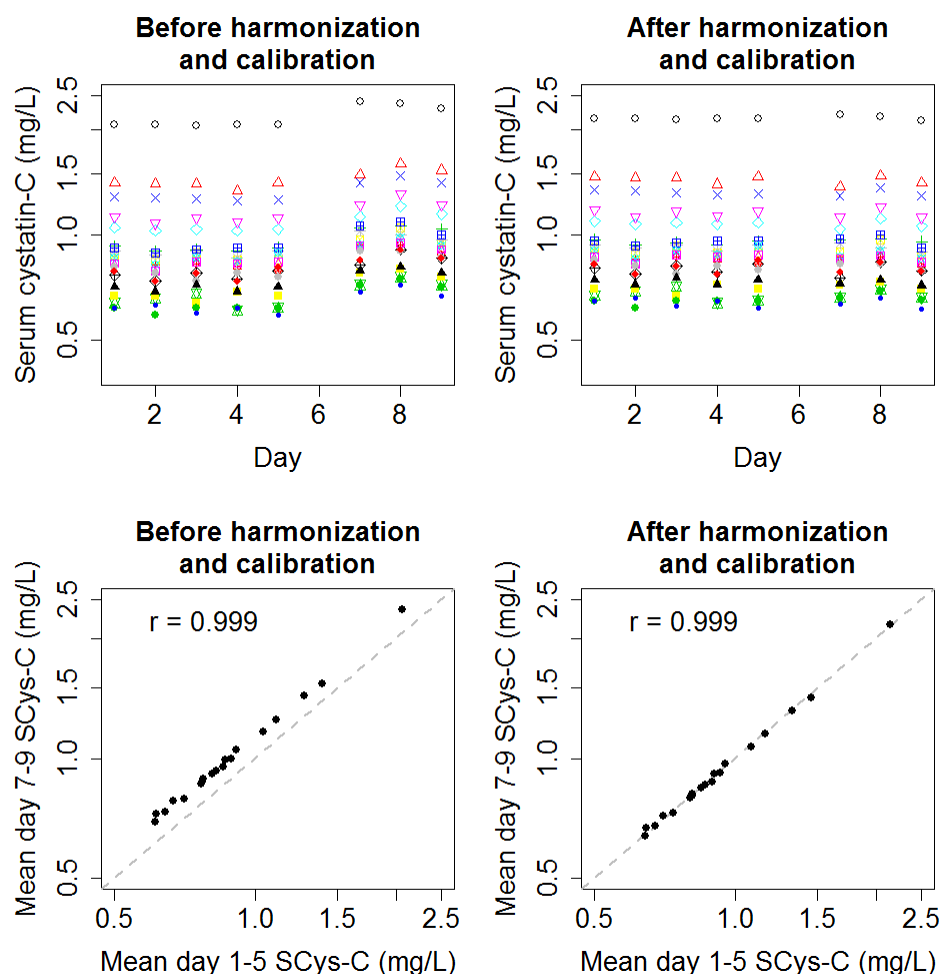
During the course of measurement, a one-time shift in serum cystatin C measurements was observed using “high” and “low” serum standards as well as the serum QC set. This shift corresponded with a change in reagent lot number and calibrators for cystatin C and can be seen in the Supplementary Methods Figure as a systematic increase in each sample in the QC set comparing the last three days on which the QC set was run to the first five days on which the QC set was run (upper and lower left panels).

Cystatin C standard ERM-DA471/IFCC was run 8 times after the shift was observed. The observed mean concentration (5.96 mg/L) was slightly higher than the expected concentration (5.49 mg/L).

We transformed our cystatin C concentrations to calibrate them with ERM-DA471/IFCC and harmonize them across the study. First, we calibrated post-shift cystatin C concentrations (including study samples and QC set samples) by multiplying each value by the ratio of the expected (5.49 mg/L) to the observed (5.961 mg/L) reference mean. After calibration, we calculated the mean cystatin C concentration for each of the 20 samples in the QC set separately for days 1-5 (before the shift) and days 7-9 (after the shift, now calibrated to ERM-DA471/IFCC). We then used the pairs of mean values of the 20 samples in the QC set to generate an equation harmonizing cystatin C concentrations measured before the shift to those measured after the shift. The regression equation was: $0.006801 + 1.037603 * (\text{pre-shift cystatin C})$

concentration). After harmonization and calibration steps, QC samples had excellent concordance ($r = 0.999$) and fell on the line of identity (lower right hand panel of the Supplementary Methods Figure). We applied this equation to all study samples measured before the cystatin C shift to harmonize these measurements to those run after the cystatin C shift.

There was no evidence of drift or shift for measurements of serum creatinine or urine albumin or creatinine (data not shown). Creatinine concentrations were within 4.5% of the NIST standard. Therefore, no transformation of these measurements was made.



Supplementary Methods References

1. de Boer IH, Sun W, Cleary PA, et al. Longitudinal changes in estimated and measured GFR in type 1 diabetes. *J Am Soc Nephrol*. Apr 2014;25(4):810-818.
2. White CA, Rule AD, Collier CP, et al. The impact of interlaboratory differences in cystatin C assay measurement on glomerular filtration rate estimation. *Clinical journal of the American Society of Nephrology : CJASN*. Sep 2011;6(9):2150-2156.
3. de Boer IH, Zelnick LR, Lin J, et al. Vitamin D and omega-3 trial to prevent and treat diabetic kidney disease: Rationale, design, and baseline characteristics. *Contemporary clinical trials*. Sep 30 2018;74:11-17.

eTable 1. Adherence Over Time, Overall and by Treatment Group

	Overall	Placebo	Active
Vitamin D intervention			
6 months	1208 (97)	566 (97)	642 (98)
1 year	1174 (95)	553 (95)	621 (94)
2 years	1079 (92)	506 (92)	573 (91)
Adherent through year 2	1194 (91)	560 (92)	634 (90)
3 years	1029 (91)	474 (91)	555 (91)
4 years	984 (91)	453 (90)	531 (91)
5 years	709 (88)	331 (89)	378 (87)
Adherent through year 5	1032 (79)	486 (80)	546 (78)
Omega-3 fatty acid intervention			
6 months	1208 (97)	600 (98)	608 (97)
1 year	1177 (95)	589 (95)	588 (95)
2 years	1077 (91)	540 (92)	537 (90)
Adherent through year 2	1193 (91)	591 (91)	602 (91)
3 years	1031 (92)	510 (91)	521 (93)
4 years	987 (91)	490 (91)	497 (91)
5 years	721 (89)	353 (89)	368 (89)
Adherent through year 5	1032 (79)	514 (79)	518 (79)

Entries are N (%), with percentages calculated among those answering the compliance question by questionnaire. Adherence at each individual time point was defined by self-report of taking at least two thirds of study medications. Adherence through year 2 is defined as reporting adherence on at least 2 of 3 questionnaires through year 2; adherence through year 5 defined as reporting adherence on at least 4 of 6 questionnaires through year 5.

eTable 2. Glucose-Lowering Medication Use Over Time

		Vitamin D		Fish oil	
	All (N = 1312)	Active (N = 703)	Placebo (N = 609)	Active (N = 659)	Placebo (N = 653)
Biguanides					
Baseline	889 (68)	469 (67)	420 (69)	446 (68)	443 (68)
Year 2	542 (55)	286 (54)	256 (56)	263 (53)	279 (57)
Year 5	638 (70)	325 (66)	313 (74)	324 (70)	314 (69)
Sulfonylureas					
Baseline	393 (30)	209 (30)	184 (30)	194 (29)	199 (30)
Year 2	240 (24)	133 (25)	107 (23)	112 (22)	128 (26)
Year 5	266 (29)	146 (30)	120 (28)	119 (26)	147 (32)
Thiazolidinediones					
Baseline	124 (9)	64 (9)	60 (10)	56 (8)	68 (10)
Year 2	60 (6)	33 (6)	27 (6)	24 (5)	36 (7)
Year 5	54 (6)	31 (6)	23 (5)	25 (5)	29 (6)
DPP-4 inhibitors					
Baseline	115 (9)	59 (8)	56 (9)	55 (8)	60 (9)
Year 2	96 (10)	55 (10)	41 (9)	43 (9)	53 (11)
Year 5	126 (14)	70 (14)	56 (13)	59 (13)	67 (15)
GLP1 receptor agonists					
Baseline	48 (4)	23 (3)	25 (4)	25 (4)	23 (4)
Year 2	16 (2)	8 (2)	8 (2)	5 (1)	11 (2)
Year 5	29 (3)	16 (3)	13 (3)	11 (2)	18 (4)
Meglitinides					
Baseline	17 (1)	6 (1)	11 (2)	6 (1)	11 (2)
Year 2	11 (1)	3 (1)	8 (2)	2 (0)	9 (2)
Year 5	11 (1)	6 (1)	5 (1)	3 (1)	8 (2)
Alpha-glucosidase inhibitors					
Baseline	7 (1)	2 (0)	5 (1)	2 (0)	5 (1)
Year 2	2 (0)	0 (0)	2 (0)	2 (0)	0 (0)
Year 5	1 (0)	1 (0)	0 (0)	0 (0)	1 (0)
Insulin					
Baseline	258 (20)	135 (19)	123 (20)	124 (19)	134 (21)
Year 2	176 (18)	105 (20)	71 (16)	87 (17)	89 (18)
Year 5	143 (16)	84 (17)	59 (14)	69 (15)	74 (16)
SGLT2 inhibitors					
Baseline	NA	NA	NA	NA	NA
Year 2	NA	NA	NA	NA	NA
Year 5	46 (5)	25 (6)	21 (4)	24 (5)	22 (5)

Percentages are calculated as a percent of nonmissing responses: N = 1312 at baseline, N = 988 at Year 2, and N = 916 at Year 5.

eTable 3. Antihypertensive Medication Use Over Time

		Vitamin D		Fish oil	
	All (N = 1312)	Active (N = 703)	Placebo (N = 609)	Active (N = 659)	Placebo (N = 653)
ACE inhibitors					
Baseline	565 (43)	311 (44)	254 (42)	284 (43)	281 (43)
Year 2	357 (36)	201 (38)	156 (34)	169 (34)	188 (38)
Year 5	379 (41)	218 (44)	161 (38)	183 (40)	196 (43)
Angiotensin II receptor blockers					
Baseline	260 (20)	131 (19)	129 (21)	127 (19)	133 (20)
Year 2	214 (22)	106 (20)	108 (24)	100 (20)	114 (23)
Year 5	270 (29)	138 (28)	132 (31)	138 (30)	132 (29)
ACEi/ARBs					
Baseline	803 (61)	428 (61)	375 (62)	400 (61)	403 (62)
Year 2	560 (57)	301 (57)	259 (57)	265 (53)	295 (60)
Year 5	637 (70)	348 (71)	289 (68)	313 (68)	324 (71)
Diuretics					
Baseline	364 (28)	198 (28)	166 (27)	193 (29)	171 (26)
Year 2	122 (12)	61 (11)	61 (13)	58 (12)	64 (13)
Year 5	153 (17)	82 (17)	71 (17)	76 (16)	77 (17)
Calcium channel blockers					
Baseline	275 (21)	151 (21)	124 (20)	133 (20)	142 (22)
Year 2	178 (18)	93 (18)	85 (19)	83 (17)	95 (19)
Year 5	241 (26)	131 (27)	110 (26)	117 (25)	124 (27)
Beta blockers					
Baseline	295 (22)	165 (23)	130 (21)	153 (23)	142 (22)
Year 2	192 (19)	115 (22)	77 (17)	98 (20)	94 (19)
Year 5	259 (28)	146 (30)	113 (27)	125 (27)	134 (30)
Mineralocorticoid receptor antagonists					
Baseline	9 (1)	5 (1)	4 (1)	2 (0)	7 (1)
Year 2	8 (1)	2 (0)	6 (1)	4 (1)	4 (1)
Year 5	12 (1)	7 (1)	5 (1)	6 (1)	6 (1)

Percentages are calculated as a percent of nonmissing responses: N = 1312 at baseline, N = 988 at Year 2, and N = 916 at Year 5.

eTable 4. Effects of Vitamin D and Omega-3 Fatty Acids on Estimated Glomerular Filtration Rate (the Primary Study Outcome) Among Participants With Available Baseline and Year 5 Measurements (N = 932*)

	Active intervention			Placebo			Difference in change from baseline**	
	N	Mean (95% CI)	Change from baseline, mean (95% CI)***	N	Mean (95% CI)	Change from baseline, mean (95% CI)***	Mean difference (95% CI)	p-value^
Vitamin D								
Baseline	495	86.2 (84.2, 88.2)	0 (Ref.)	437	85.4 (83.5, 87.3)	0 (Ref.)	0 (Ref.)	
Year 2	443	81.3 (79.4, 83.2)	-4.9 (-6.0, -3.7)	389	79.4 (77.3, 81.5)	-6.1 (-7.2, -4.9)	1.18 (-0.59, 2.95)	
Year 5	495	74.2 (72.3, 76.2)	-12.0 (-13.3, -10.7)	437	72.5 (70.5, 74.5)	-12.8 (-14.2, -11.5)	0.87 (-0.83, 2.58)	0.32
Omega-3 fatty acids								
Baseline	470	86.0 (84.2, 87.9)	0 (Ref.)	462	85.6 (83.5, 87.6)	0 (Ref.)	0 (Ref.)	
Year 2	424	79.9 (78.0, 81.8)	-5.9 (-7.1, -4.7)	408	80.9 (78.8, 83.0)	-5.0 (-6.1, -3.8)	-0.93 (-2.66, 0.81)	
Year 5	470	73.7 (71.8, 75.6)	-12.4 (-13.7, -11.0)	462	73.2 (71.1, 75.3)	-12.4 (-13.8, -11.0)	0.09 (-1.61, 1.80)	0.92

Mean and SD values are in mL/min/1.73m².

* Of 1,312 enrolled participants, 1,308 donated a baseline blood sample, and 932 of these also donated a year 5 blood sample and were included in this analysis. 376 participants who donated a blood sample at baseline but not year 5 were excluded, as were 4 participants who did not donate a baseline blood sample (2 of whom donated a blood sample at year 5).

** Modeled difference, with positive values indicating higher eGFR at year 5 (less loss of eGFR from baseline) among participants assigned to the active intervention compared to those assigned to the corresponding placebo intervention. From a linear mixed model that includes adjustment for age, sex, and baseline urine ACR and accounts for missing data using multiple imputation.

*** Change from baseline summarized over all participants using multiple imputation.

^ p-value tests the difference in change in eGFR from baseline to Year 5.

eTable 5. Effects of Vitamin D and Omega-3 Fatty Acids on Estimated Glomerular Filtration Rate (the Primary Study Outcome) Among Adherent Participants

	Active intervention			Placebo			Difference in change from baseline*	
	N	Mean (95% CI)	Change from baseline, mean (95% CI)**	N	Mean (95% CI)	Change from baseline, mean (95% CI)**	Mean difference (95% CI)	p-value***
Vitamin D								
Baseline	544	86.2 (84.3, 88.1)	0 (Ref.)	485	85.8 (83.9, 87.6)	0 (Ref.)	0 (Ref.)	
Year 2	482	80.9 (79.1, 82.8)	-5.3 (-6.3, -4.3)	415	80.0 (77.9, 82.0)	-6.0 (-7.1, -5.0)	0.73 (-0.89, 2.36)	
Year 5	461	74.4 (72.4, 76.4)	-12.0 (-13.2, -10.7)	404	72.6 (70.6, 74.7)	-12.8 (-14.1, -11.5)	0.89 (-0.74, 2.52)	0.28
Omega-3 fatty acids								
Baseline	517	86.2 (84.4, 88.0)	0 (Ref.)	513	85.5 (83.6, 87.5)	0 (Ref.)	0 (Ref.)	
Year 2	457	79.9 (78.0, 81.7)	-5.8 (-6.9, -4.8)	441	80.9 (78.8, 83.0)	-5.4 (-6.5, -4.3)	-0.48 (-2.10, 1.14)	
Year 5	438	73.8 (71.8, 75.8)	-12.1 (-13.3, -10.8)	426	73.3 (71.2, 75.4)	-12.5 (-13.8, -11.3)	0.42 (-1.22, 2.06)	0.61

Mean and SD values are in mL/min/1.73m².

* Modeled difference, with positive values indicating higher eGFR at year 5 (less loss of eGFR from baseline) among participants assigned to the active intervention compared to those assigned to the corresponding placebo intervention. From a linear mixed model that includes adjustment for age, sex, and baseline urine ACR and accounts for missing data using multiple imputation.

** Change from baseline summarized over all participants using multiple imputation.

***p-value tests the difference in change in eGFR from baseline to Year 5.

Adherent population defined as all participants who reported taking ≥ 2/3 of study drug at least 4 times (questioned at 6 months, 1 year, 2 years, 3 years, 4 years, and 5 years): N = 1032 for Vitamin D intervention, N = 1032 for fish oil intervention.

eTable 6. Effects of Vitamin D and Omega-3 Fatty Acids on Urine Albumin-Creatinine Ratio

	Active intervention			Placebo			Difference in change from baseline*	
	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	Ratio of change from baseline, active to placebo (95% CI)	p-value***
Vitamin D								
Baseline	702	3.3 (3.0, 3.8)	1.0 (Ref.)	609	3.2 (2.8, 3.6)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	529	5.5 (4.8, 6.5)	1.79 (1.60, 2.01)	463	4.7 (4.0, 5.5)	1.57 (1.41, 1.76)	1.14 (0.96, 1.37)	
Year 5	505	9.3 (8.1, 10.6)	2.97 (2.66, 3.32)	440	9.2 (8.0, 10.5)	3.02 (2.70, 3.38)	0.99 (0.84, 1.17)	0.90
Omega-3 fatty acids								
Baseline	658	3.3 (2.9, 3.8)	1.0 (Ref.)	653	3.2 (2.8, 3.6)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	502	5.2 (4.5, 6.2)	1.66 (1.48, 1.86)	490	5.0 (4.3, 5.8)	1.71 (1.53, 1.91)	0.97 (0.81, 1.16)	
Year 5	478	9.1 (7.9, 10.5)	2.94 (2.62, 3.30)	467	9.4 (8.2, 10.6)	3.05 (2.74, 3.40)	0.96 (0.81, 1.14)	0.64

Mean and SD values are in mg albumin per g creatinine.

* From a linear mixed model that includes adjustment for age and sex and accounts for missing data using multiple imputation

**Change from baseline summarized over all participants using multiple imputation.

***p-value tests the difference in change in uACR between baseline and Year 5.

eTable 7. Effects of Vitamin D and Omega-3 Fatty Acids on Urine Albumin-Creatinine Ratio Among Participants With Available Baseline and Year 5 Measurements (N = 944*)

	Active intervention			Placebo			Difference in change from baseline**	
	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)***	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)***	Ratio of change from baseline, active to placebo (95% CI)	p-value^
Vitamin D								
Baseline	504	3.0 (2.6, 3.5)	1.0 (Ref.)	440	3.1 (2.6, 3.5)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	449	5.2 (4.5, 6.1)	1.80 (1.58, 2.05)	396	4.5 (3.8, 5.3)	1.51 (1.33, 1.71)	1.20 (1.00, 1.44)	
Year 5	504	9.3 (8.1, 10.7)	3.08 (2.70, 3.50)	440	9.2 (8.0, 10.5)	3.00 (2.65, 3.41)	1.03 (0.86, 1.22)	0.77
Omega-3 fatty acids								
Baseline	477	3.1 (2.7, 3.6)	1.0 (Ref.)	467	3.0 (2.6, 3.4)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	433	5.0 (4.2, 5.9)	1.62 (1.43, 1.84)	412	4.7 (4.0, 5.5)	1.69 (1.49, 1.92)	0.96 (0.80, 1.15)	
Year 5	477	9.1 (7.9, 10.5)	2.94 (2.58, 3.36)	467	9.4 (8.2, 10.6)	3.15 (2.77, 3.57)	0.93 (0.78, 1.11)	0.44

Mean and SD values are in mg albumin per g creatinine.

* Of 1,312 enrolled participants, 1,311 donated a baseline urine sample, and 991 of these also donated a year 5 urine sample and were included in this analysis. 320 participants who donated a urine sample at baseline but not year 5 were excluded, as was 1 participant who did not donate a baseline urine sample.

**From a linear mixed model that includes adjustment for age and sex and accounts for missing data using multiple imputation

***Change from baseline summarized over all participants using multiple imputation.

^p-value tests the difference in change between baseline and Year 5.

eTable 8. Effects of Vitamin D and Omega-3 Fatty Acids on Urine Albumin-Creatinine Ratio Among Adherent Participants

	Active intervention			Placebo			Difference in change from baseline*	
	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	Ratio of change from baseline, active to placebo (95% CI)	p-value***
Vitamin D								
Baseline	545	3.1 (2.7, 3.5)	1.0 (Ref.)	486	3.1 (2.7, 3.6)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	481	5.4 (4.6, 6.3)	1.80 (1.58, 2.05)	420	4.4 (3.7, 5.2)	1.51 (1.34, 1.70)	1.20 (1.00, 1.43)	
Year 5	467	9.2 (8.0, 10.6)	3.03 (2.68, 3.44)	407	9.1 (7.9, 10.5)	2.99 (2.64, 3.37)	1.02 (0.85, 1.22)	0.87
Omega-3 fatty acids								
Baseline	517	3.1 (2.7, 3.5)	1.0 (Ref.)	514	3.1 (2.7, 3.5)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	461	4.9 (4.2, 5.8)	1.63 (1.43, 1.86)	441	4.8 (4.2, 5.7)	1.70 (1.51, 1.91)	0.96 (0.80, 1.16)	
Year 5	443	9.2 (7.9, 10.7)	3.03 (2.67, 3.44)	430	9.1 (8.0, 10.4)	3.04 (2.70, 3.43)	0.99 (0.83, 1.19)	0.94

Mean and SD values are in mg albumin per g creatinine.

*From a linear mixed model that includes adjustment for age and sex and accounts for missing data using multiple imputation

**Change from baseline summarized over all participants using multiple imputation.

***p-value tests the difference in change between baseline and Year 5.

Adherent population defined as all participants who reported taking $\geq 2/3$ of study drug at least 4 times (questioned at 6 months, 1 year, 2 years, 3 years, 4 years, and 5 years): N = 1032 for Vitamin D intervention, N = 1032 for fish oil intervention.

eTable 9. Effects of Vitamin D and Omega-3 Fatty Acids on Urine Albumin-Creatinine Ratio (the Secondary Study Outcome) in the VITAL DKD Ancillary Study, Excluding Visits in Which Participants Reported UTI Symptoms

	Active intervention			Placebo			Difference in change from baseline*	
	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	N	Geometric mean (95% CI)	Change from baseline (ratio), geometric mean (95% CI)**	Ratio of change from baseline, active to placebo (95% CI)	p-value***
Vitamin D								
Baseline	680	3.3 (2.9, 3.7)	1.0 (Ref.)	582	3.1 (2.7, 3.5)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	510	5.5 (4.7, 6.4)	1.77 (1.59, 1.98)	449	4.5 (3.9, 5.3)	1.52 (1.37, 1.70)	1.16 (0.97, 1.39)	
Year 5	487	9.1 (7.9, 10.4)	2.94 (2.63, 3.28)	422	9.1 (7.9, 10.4)	2.98 (2.67, 3.32)	0.99 (0.84, 1.17)	0.90
Omega-3 fatty acids								
Baseline	635	3.3 (2.9, 3.7)	1.0 (Ref.)	627	3.1 (2.8, 3.6)	1.0 (Ref.)	1.0 (Ref.)	
Year 2	482	5.0 (4.2, 5.9)	1.62 (1.44, 1.82)	477	5.0 (4.3, 5.8)	1.69 (1.52, 1.88)	0.96 (0.80, 1.15)	
Year 5	460	9.2 (8.0, 10.6)	2.91 (2.60, 3.25)	449	9.0 (7.9, 10.2)	3.01 (2.70, 3.35)	0.98 (0.83, 1.16)	0.80

Mean and SD values are in mg albumin per g creatinine.

*From a linear mixed model that includes adjustment for age and sex and accounts for missing data using multiple imputation

**Change from baseline summarized over all participants using multiple imputation.

***p-value tests the difference in change between baseline and Year 5.

eTable 10. Effects of Vitamin D and Omega-3 Fatty Acids on Exploratory and Post-Hoc eGFR Outcomes

	Active intervention		Placebo		Difference in IR per 100 person- years (95% CI)	Hazard ratio, active v placebo (95% CI)	p-value
	N events	Incidence rate per 100 person- years (95% CI)	N events	Incidence rate per 100 person- years (95% CI)			
Vitamin D intervention							
≥ 40% decline in eGFR or kidney failure during study	46	1.7 (1.2, 2.2)	39	1.7 (1.2, 2.2)	0.1 (-0.7, 0.8)	1.03 (0.68, 1.58)	0.88
≥ 30% decline in eGFR, kidney failure during study, or death	125	3.4 (2.8, 4.0)	130	4.2 (3.5, 4.8)	-0.7 (-1.6, 0.2)	0.82 (0.64, 1.05)	0.12
≥ 30% decline in eGFR or kidney failure during study	87	3.4 (2.7, 4.1)	92	4.1 (3.3, 4.8)	-0.7 (-1.7, 0.3)	0.82 (0.61, 1.09)	0.17
≥ 30% decline in eGFR	83	3.2 (2.5, 3.9)	91	4.0 (3.3, 4.8)	-0.8 (-1.8, 0.2)	0.79 (0.59, 1.06)	0.12
Omega-3 intervention							
≥ 40% decline in eGFR or kidney failure during study	44	1.8 (1.3, 2.3)	41	1.7 (1.2, 2.2)	0.1 (-0.6, 0.8)	1.07 (0.70, 1.63)	0.77
≥ 30% decline in eGFR, kidney failure during study, or death	125	3.7 (3.1, 4.3)	130	3.8 (3.2, 4.5)	-0.1 (-1.0, 0.8)	0.96 (0.75, 1.23)	0.77
≥ 30% decline in eGFR or kidney failure during study	85	3.5 (2.8, 4.2)	94	3.9 (3.2, 4.6)	-0.4 (-1.4, 0.6)	0.89 (0.66, 1.19)	0.44
≥ 30% decline in eGFR	81	3.3 (2.6, 4.0)	93	3.9 (3.1, 4.6)	-0.5 (-1.5, 0.5)	0.86 (0.64, 1.15)	0.31

Hazard ratios come from Cox regression model; eGFR outcomes exclude N = 4 participants who were missing baseline eGFR and uACR outcome excludes N = 1 participant who was missing baseline uACR. P-value tests the null hypothesis that HR = 1. These analyses were not pre-specified and are included here as post-hoc analyses.

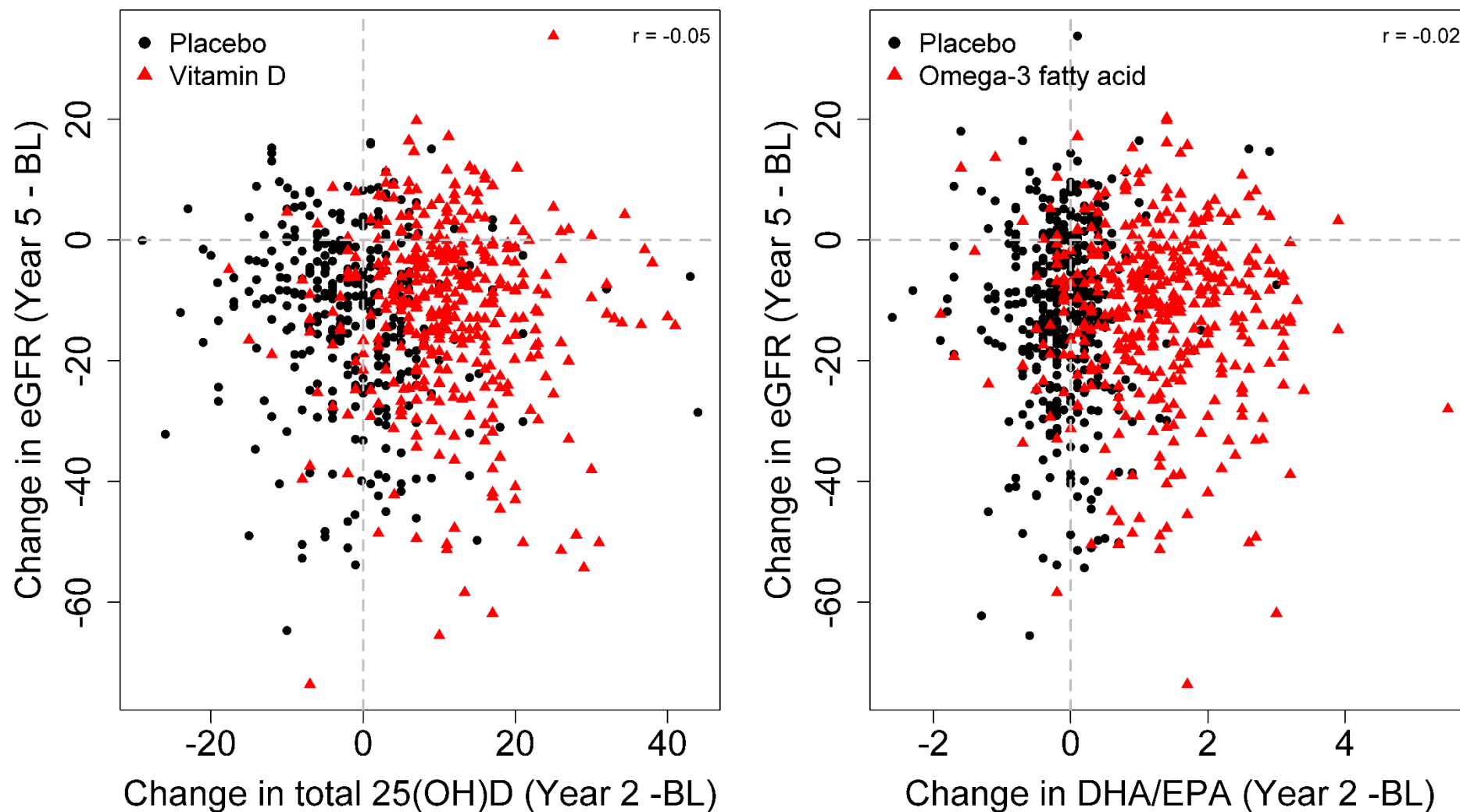
eTable 11. Adverse Effects

	Vitamin D assignment		Omega-3 fatty acid assignment	
Outcome	Vitamin D (N=703)	Placebo (N=609)	Omega-3 fatty acids (N=659)	Placebo (N=653)
<i>Monitored safety conditions</i>				
Hypercalcemia	9	10	14	5
Kidney stones	32	26	31	27
Parathyroid condition	3	2	4	1
Gastrointestinal bleeding	19	26	28	17
Blood in urine	55	61	57	59
Easy bruising	197	188	190	195
Frequent nosebleeds	23	30	24	29
<i>Other symptoms and side effects</i>				
Stomach upset or pain	277	262	283	256
Nausea	216	223	235	204
Constipation	339	291	317	313
Diarrhea	362	348	352	358
Skin rash	197	165	186	176
Bad taste in mouth	165	145	151	159
Increased burping	138	109	131	116

Cell contents are numbers of participants reporting the outcome on at least one occasion.

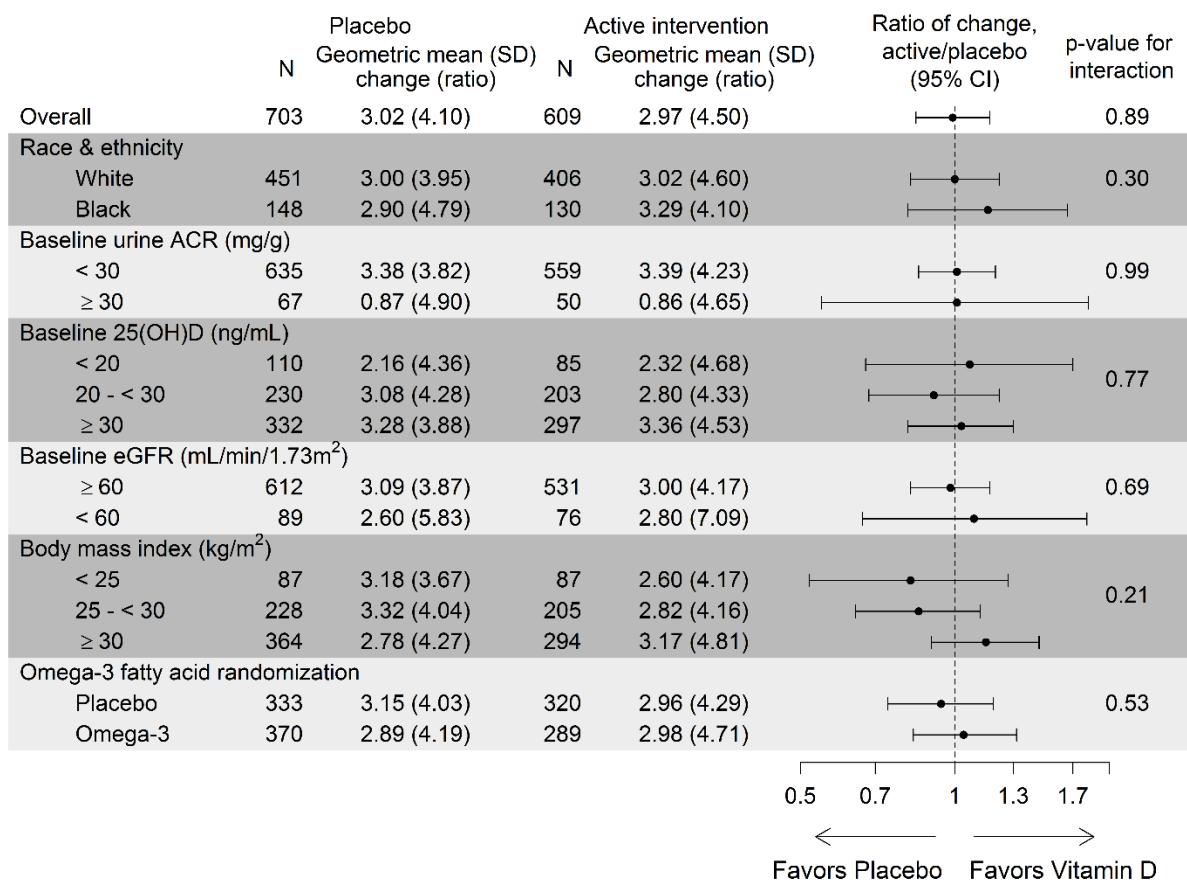
eFigure 1. Correlations of Changes in Intermediate Markers of Supplement Intake With Changes in Estimated Glomerular Filtration Rate

Changes in serum 25-hydroxyvitamin D concentration and omega-3 index are evaluated from baseline to year 2. Changes in eGFR are evaluated from baseline to year 5. Only participants with all available data were included in these analyses.



eFigure 2. Effects of Vitamin D Versus Placebo on Change in Urine Albumin-Creatinine Ratio (ACR) Among Participant Subgroups

Estimates are difference in change in urine ACR from baseline to year 5, comparing the active treatment assignment to year 5, adjusted for age and sex.



Estimates are difference in change in urine ACR from baseline to year 5, comparing the active treatment assignment to year 5, adjusted for age and sex.

0.5 0.7 1 1.3 1.7

← Favors Placebo Favors Omega-3 →